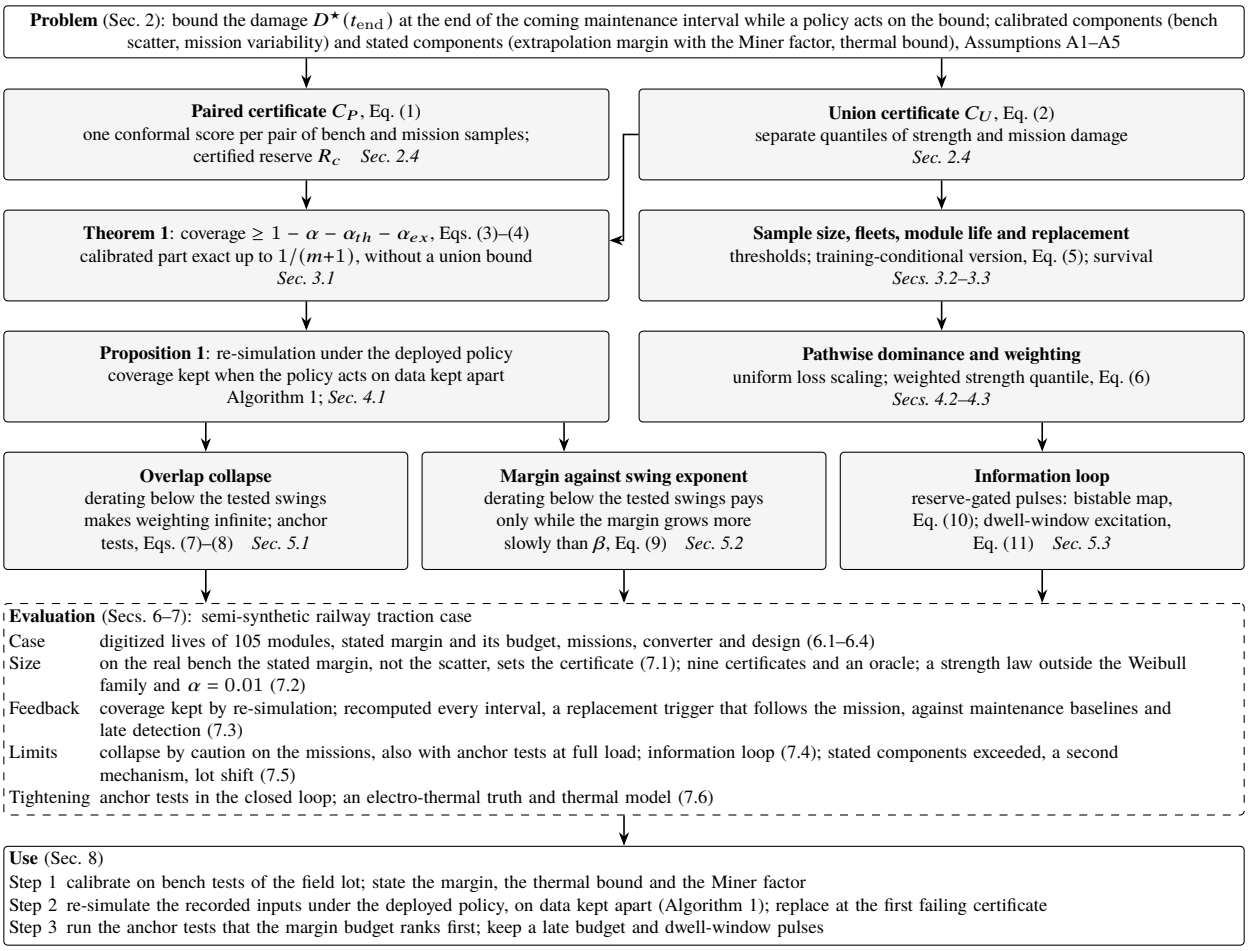
Problem (Sec. 2): bound the damage $D^\star(t_{\mathrm{end}})$ at the end of the coming maintenance interval while a policy acts on the bound; calibrated components (bench scatter, mission variability) and stated components (extrapolation margin with the Miner factor, thermal bound), Assumptions A1–A5
Paired certificate C_P, Eq. (1)
one conformal score per pair of bench and mission samples;
certified reserve R_c *Sec. 2.4*
Union certificate C_U, Eq. (2)
separate quantiles of strength and mission damage
Sec. 2.4
Theorem 1: coverage $\geq 1 - \alpha - \alpha_{th} - \alpha_{ex}$, Eqs. (3)–(4)
calibrated part exact up to $1/(m+1)$, without a union bound
Sec. 3.1
Sample size, fleets, module life and replacement
thresholds; training-conditional version, Eq. (5); survival
Secs. 3.2–3.3
Proposition 1: re-simulation under the deployed policy
coverage kept when the policy acts on data kept apart
Algorithm 1; *Sec. 4.1*
Pathwise dominance and weighting
uniform loss scaling; weighted strength quantile, Eq. (6)
Secs. 4.2–4.3
Overlap collapse
derating below the tested swings
makes weighting infinite; anchor
tests, Eqs. (7)–(8) *Sec. 5.1*
Margin against swing exponent
derating below the tested swings pays
only while the margin grows more
slowly than β, Eq. (9) *Sec. 5.2*
Information loop
reserve-gated pulses: bistable map,
Eq. (10); dwell-window excitation,
Eq. (11) *Sec. 5.3*
Evaluation (Secs. 6–7): semi-synthetic railway traction case
Case digitized lives of 105 modules, stated margin and its budget, missions, converter and design (6.1–6.4)
Size on the real bench the stated margin, not the scatter, sets the certificate (7.1); nine certificates and an oracle; a strength law outside the Weibull family and $\alpha = 0.01$ (7.2)
Feedback coverage kept by re-simulation; recomputed every interval, a replacement trigger that follows the mission, against maintenance baselines and late detection (7.3)
Limits collapse by caution on the missions, also with anchor tests at full load; information loop (7.4); stated components exceeded, a second mechanism, lot shift (7.5)
Tightening anchor tests in the closed loop; an electro-thermal truth and thermal model (7.6)
Use (Sec. 8)
Step 1 calibrate on bench tests of the field lot; state the margin, the thermal bound and the Miner factor
Step 2 re-simulate the recorded inputs under the deployed policy, on data kept apart (Algorithm 1); replace at the first failing certificate
Step 3 run the anchor tests that the margin budget ranks first; keep a late budget and dwell-window pulses